

CMP682 — Final Project

# MissVARPath

*Missense Variant Pathogenicity Prediction*

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# Three questions

Every missense variant goes through a battery of in-silico predictors — PolyPhen-2, SIFT, REVEL, CADD, AlphaMissense, DITTO. Each disagrees with the others; most were trained on overlapping ClinVar data — the same database we use to label our targets.

- 1.** Can a meta-classifier do better than any single predictor by combining them?
- 2.** How much of any improvement is data-circularity — predictors trained on ClinVar, evaluated on ClinVar?
- 3.** Can we predict pathogenicity from raw data alone — without using any pre-existing predictor's score?

# Data

**21,872**

ClinVar / OpenCRAVAT missense variants

**5,468 × 4**

perfectly balanced 4-class — Benign / Likely-benign / Likely-pathogenic / Pathogenic

**777 → 208**

raw schema → numerical features after preprocessing

**5,428**

strict-VUS rows in the curated pro-set

*Four task definitions used in the project*

## **4-class**

original ClinVar label space

## **2-class**

Likely-\* collapsed into definitive labels → 10,936 / 10,936

## **3-class (with VUS)**

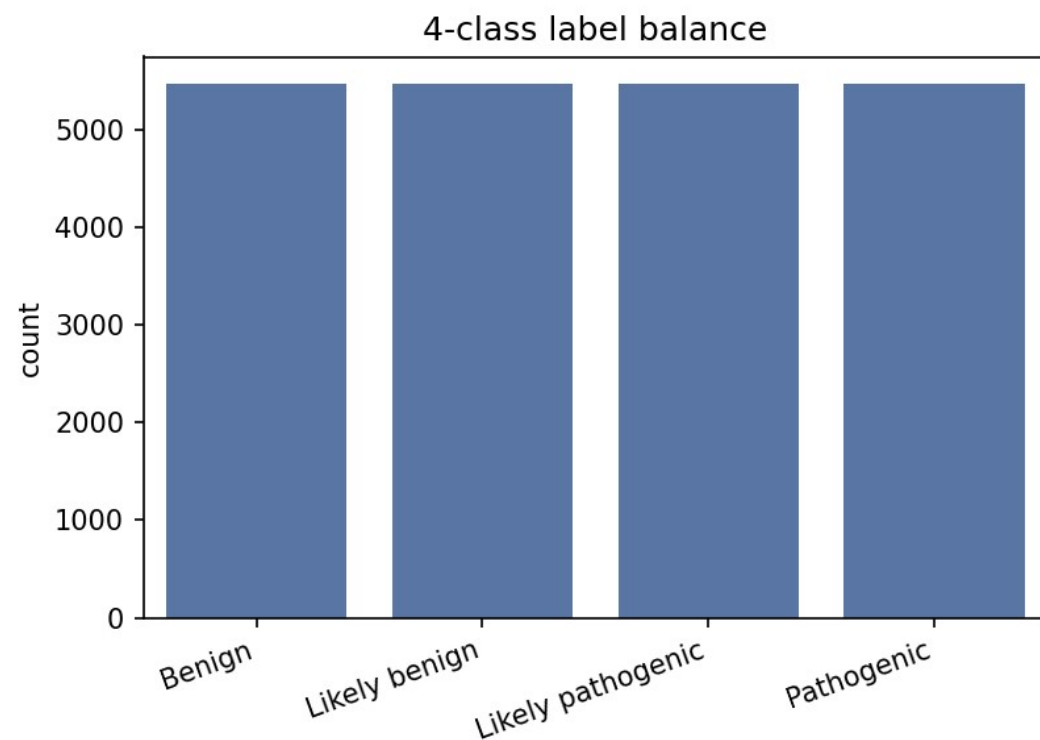
Benign-side / Pathogenic-side / VUS

## **5-class (with VUS)**

all four ClinVar labels + VUS — essentially balanced

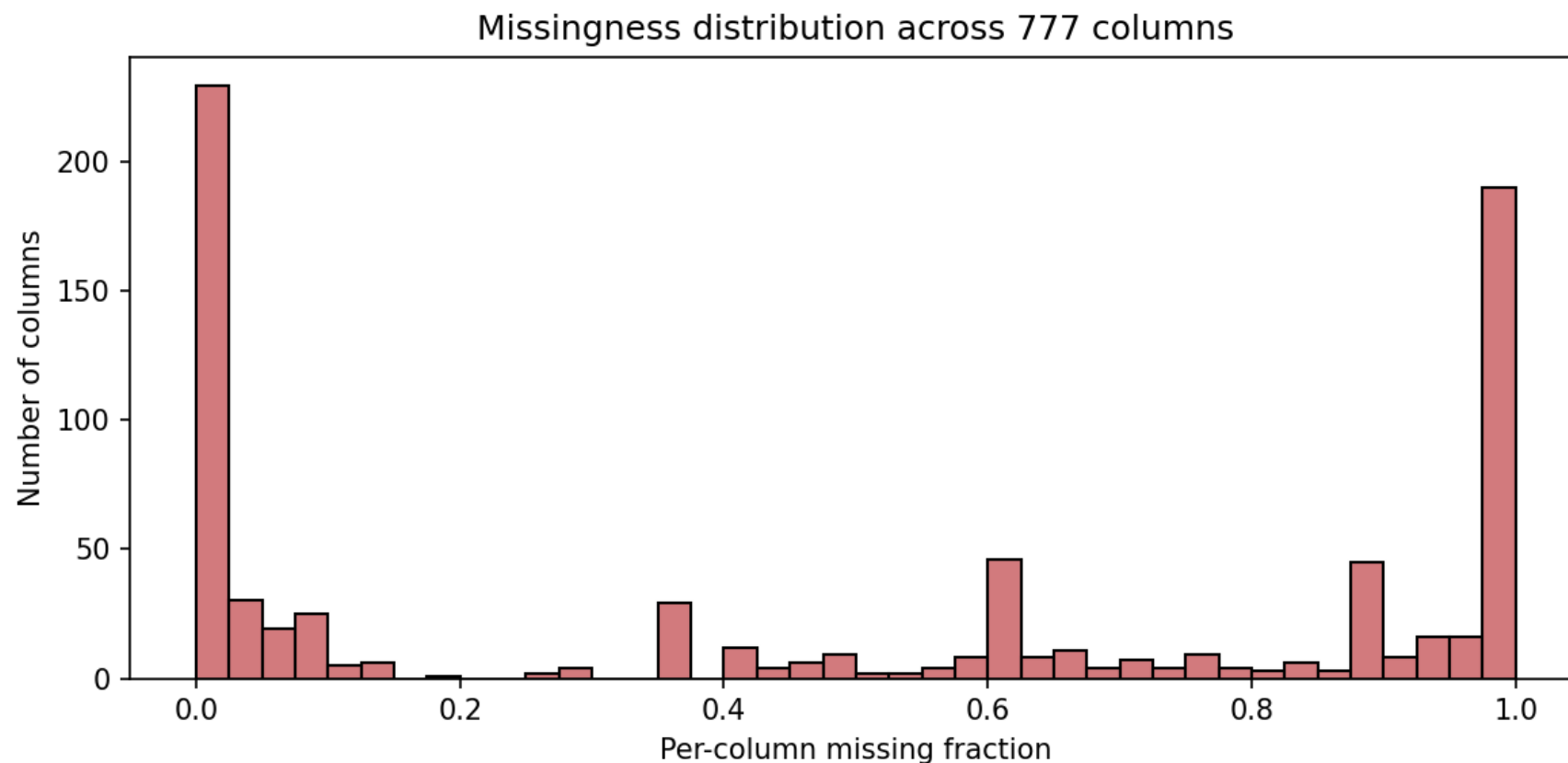
# Label balance — by design

*5,468 rows per ClinVar class in the 4-class space; 10,936 / 10,936 in the 2-class collapse. Any class-imbalance effect is gone by construction.*



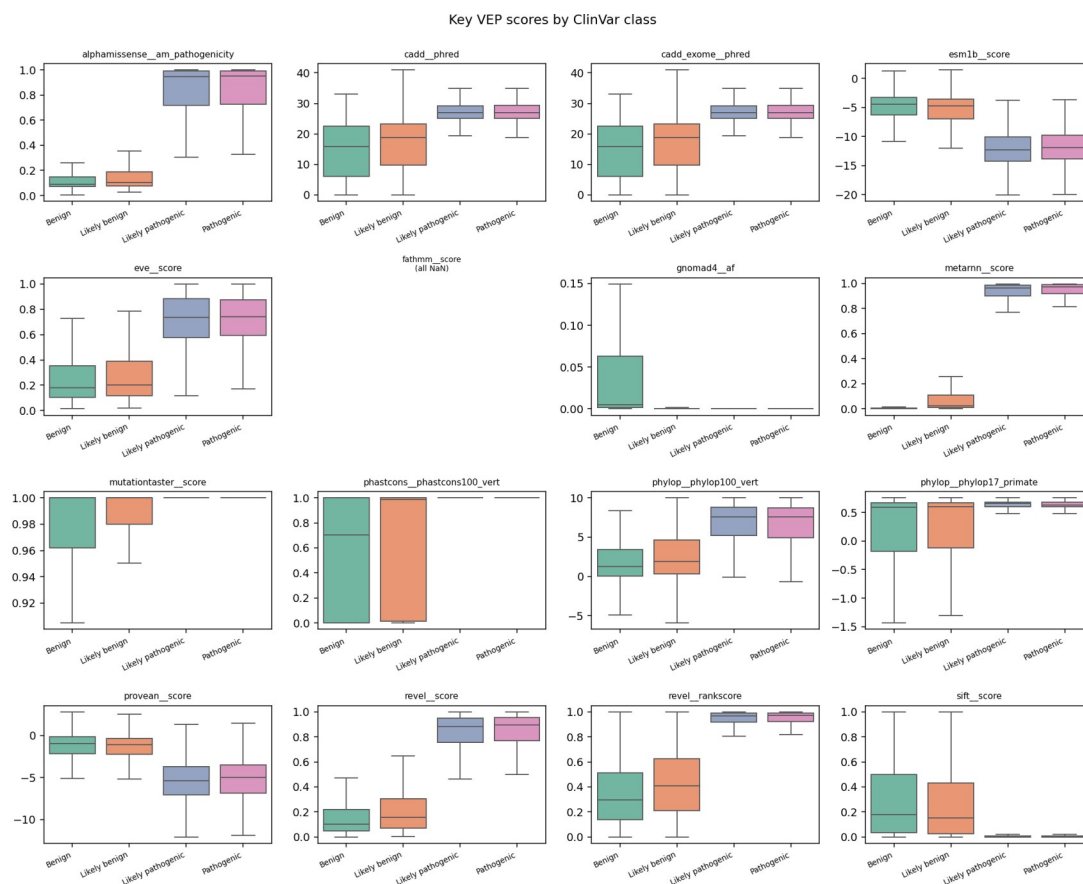
# Missingness across the 777-column raw schema

*Per-column missing fraction. Everything above the 50% threshold is dropped during preprocessing — 134 columns lost that way. 208 numerical features survive.*



# Do the predictor scores actually separate classes?

Boxplots of headline VEP scores stratified by the 4-class label. DITTO, MetaRNN, REVEL, AlphaMissense all separate cleanly; gnomAD AF runs the other way — high frequency means benign (the ACMG BS1 rule).



# Methods

*Model suite — 10 estimators*

## Classical / linear

KNN  
NearestCentroid  
CosineSimilarity (custom)  
DecisionTree  
LDA  
QDA  
LinearSVC  
RidgeClassifier  
SGDClassifier

## Ensembles

HistGradientBoosting

*Evaluation regimes — 5 cuts*

|                        |                                                         |
|------------------------|---------------------------------------------------------|
| <b>Canonical</b>       | stratified 80/20 holdout + 5-fold CV                    |
| <b>Augmented</b>       | + 196 k-mer + 10 BLAST locus-neighbour features         |
| <b>Gene-stratified</b> | no gene appears in both train and test folds            |
| <b>No-VEP / VUS</b>    | drop every learned-predictor + conservation score       |
| <b>Raw-only</b>        | keep only population AF, position, sequence composition |

*RANDOM\_STATE = 42 throughout. Per-model grid-search tuning by 5-fold CV macro-F1 on the training portion; tuning performed on the 4-class task, hyperparameters carried over to the binary task.*

# Headline numbers

4-class macro-F1

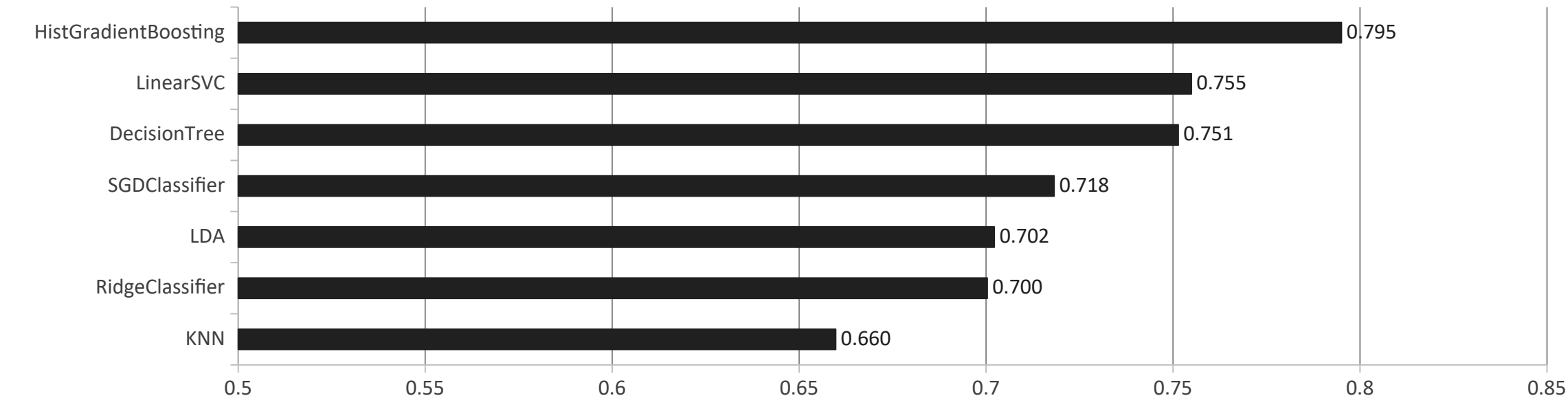
0.795

HistGradientBoosting · 21,872 variants · 208 features

2-class macro-F1

0.987

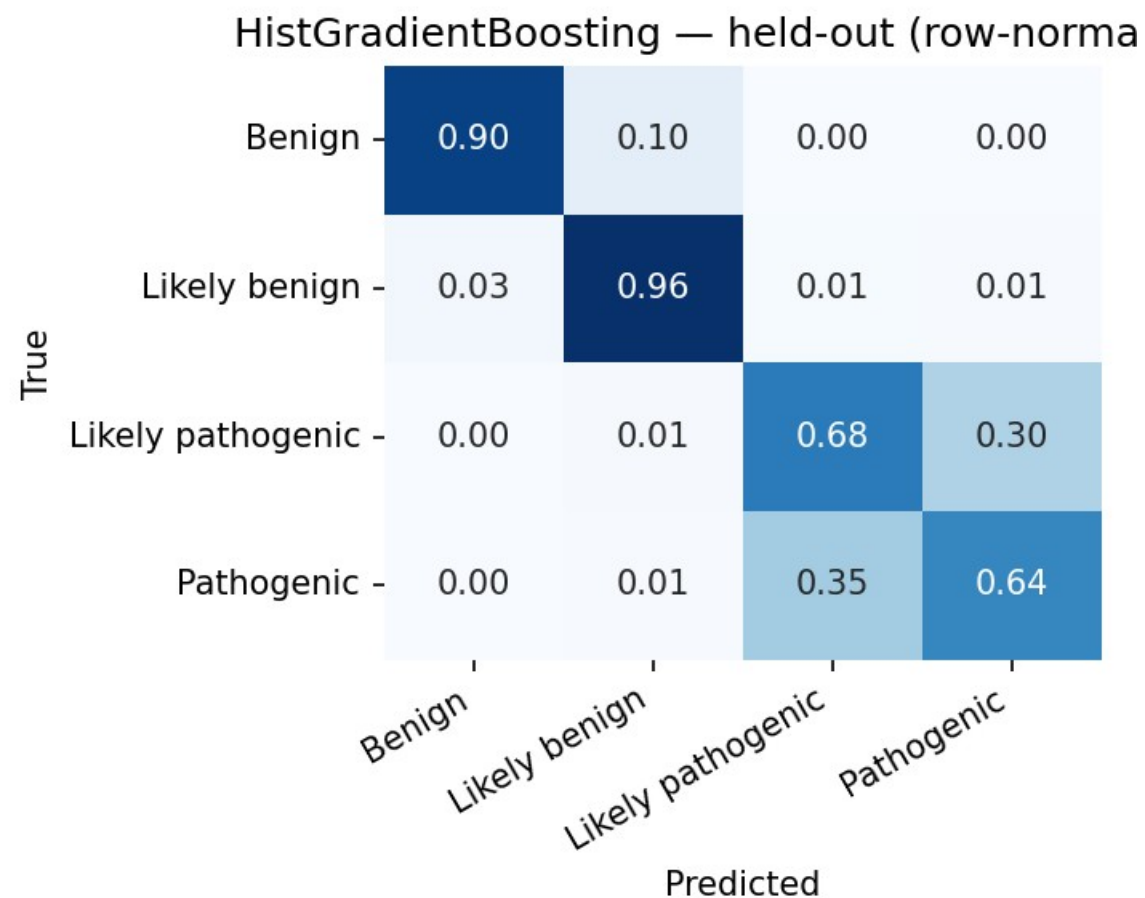
HistGradientBoosting · 10,936 / 10,936 collapse





# Where the error lives — 4-class confusion matrix

*Held-out, row-normalized. Definitive Benign / Pathogenic call near-perfect (>0.93 / 0.79). Residual difficulty is the Pathogenic  $\leftrightarrow$  Likely-pathogenic boundary.*



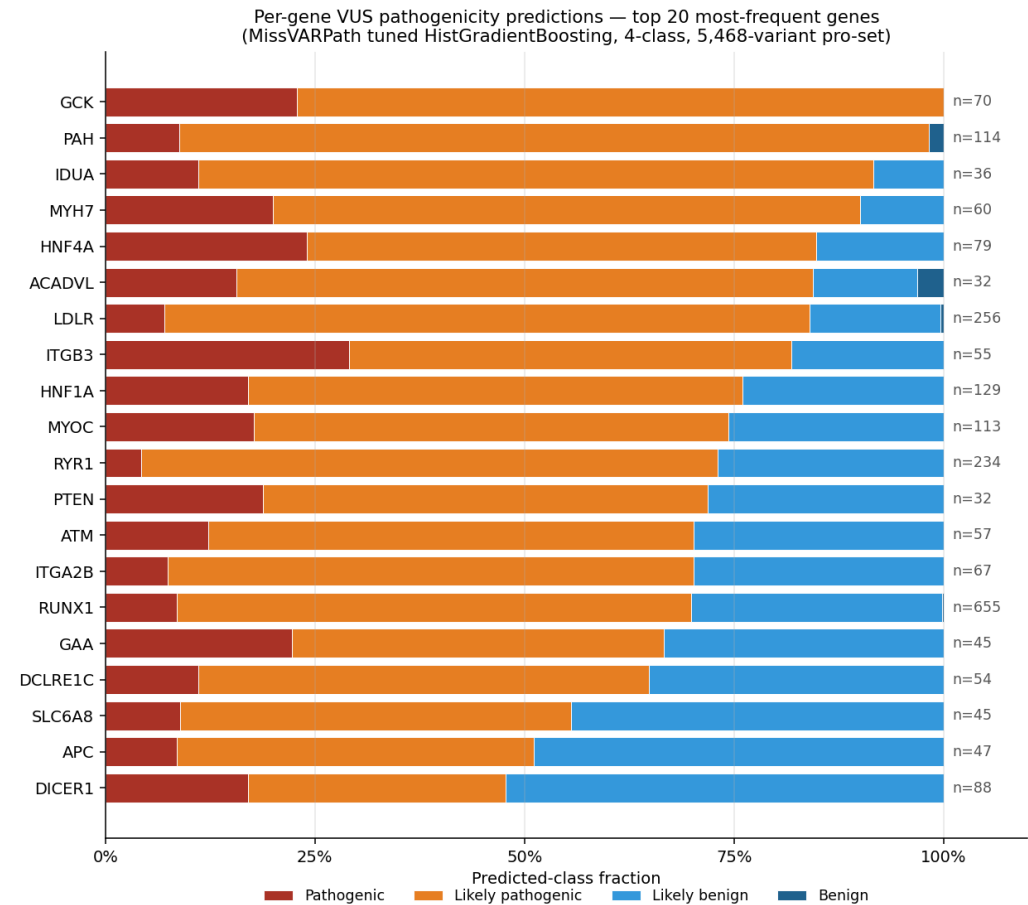
# Stress tests — bounding the data-circularity

| Regime                                   | 4-class F1   | 2-class F1   |
|------------------------------------------|--------------|--------------|
| <b>Canonical tuned (headline)</b>        | <b>0.795</b> | <b>0.987</b> |
| Augmented (k-mer + BLAST)                | 0.795        | 0.990        |
| Gene-stratified CV                       | 0.766        | 0.987        |
| No-VEP, k-fold                           | 0.773        | 0.976        |
| No-VEP, gene-stratified                  | 0.761        | —            |
| Raw-only (no predictor, no conservation) | 0.721        | 0.934        |
| DITTO removed                            | 0.791        | —            |

*Graceful degradation across regimes: HistGB loses 0.027 to gene-stratified CV, 0.022 to no-VEP, 0.074 to raw-only. The 2-class task is essentially insensitive to all stresses.*

# Real VUS deployment — per-gene predictions

Top-20 most-frequent genes in the 5,468-variant pro-set, stacked by 4-class prediction. GCK 100%, PAH 99%, MYH7 90%, LDLR 84% pathogenic-side.



# VUS deployment — the four numbers

*Applying the tuned HistGradientBoosting to 5,468 strict-VUS variants annotated through the same pipeline.*

**62% / 38%**

pathogenic-side / benign-side split on 2-class head

**80%**

of predictions have max probability > 0.9

**87%**

of 4-class predictions hedge into Likely-\* — only 13% definitive

**95.4%**

cross-task agreement (4-class collapsed vs. 2-class head)

# VUS as a class

*With VUS added as a labelled class*

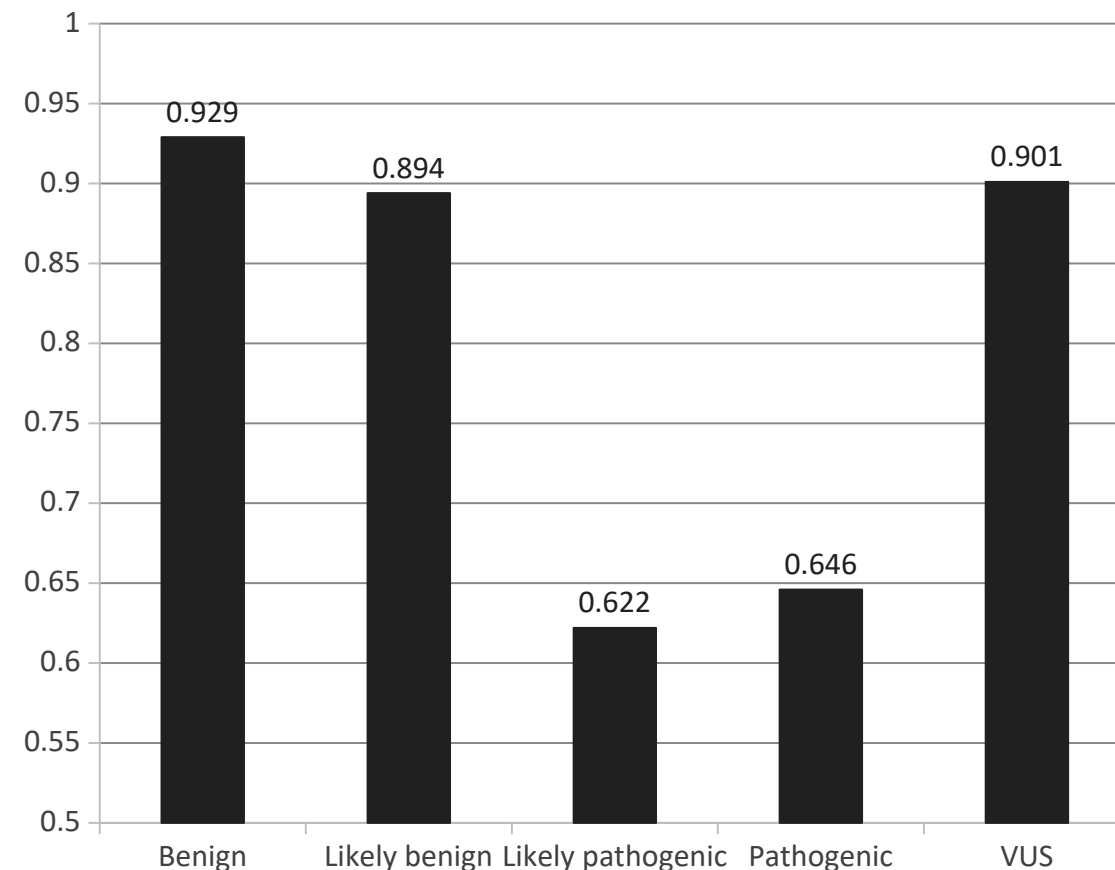
2-class no-VUS **0.987**

3-class with VUS **0.946**

4-class no-VUS **0.795**

5-class with VUS **0.798**

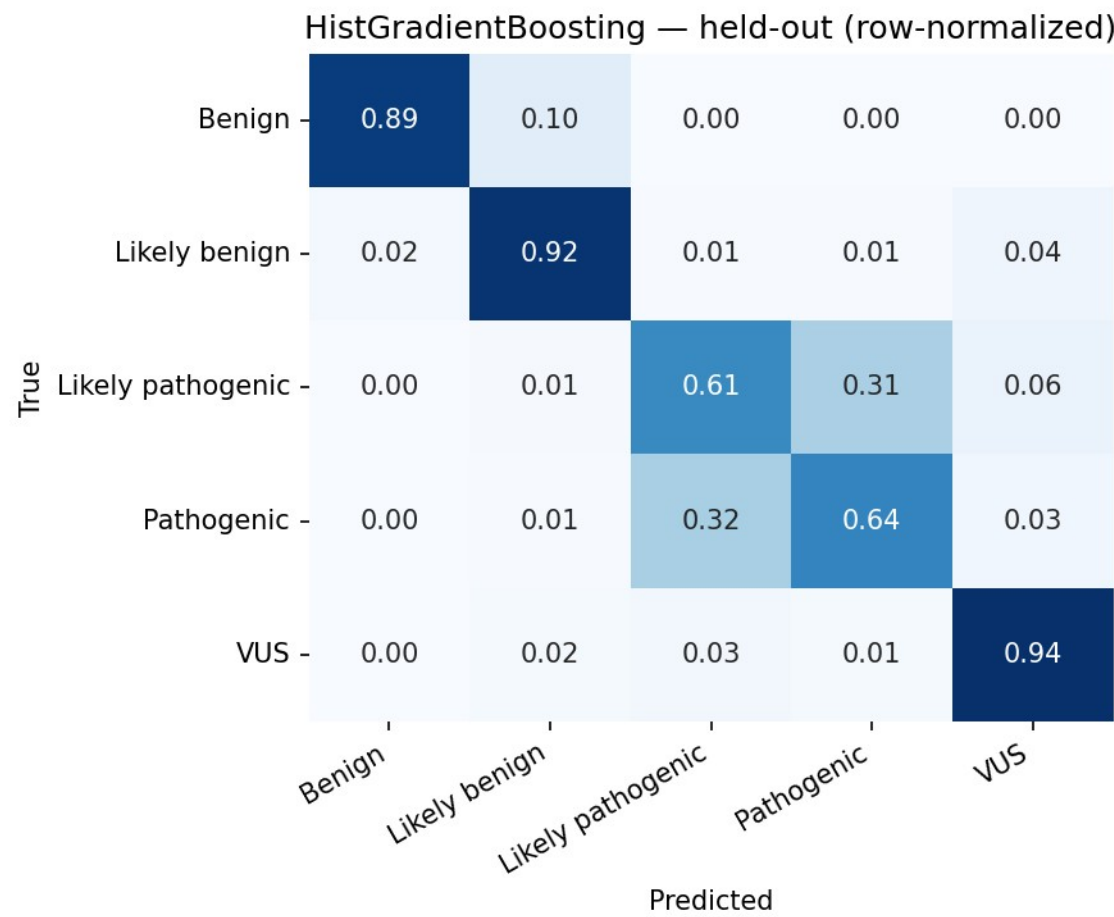
*Adding VUS as a 5th class does NOT hurt overall macro-F1 — it slightly improves it. MCC improves by +0.022.*



*VUS reaches  $F1 = 0.901$  — between Benign and Likely-benign in difficulty. One of the cleanest classes to recognise.*

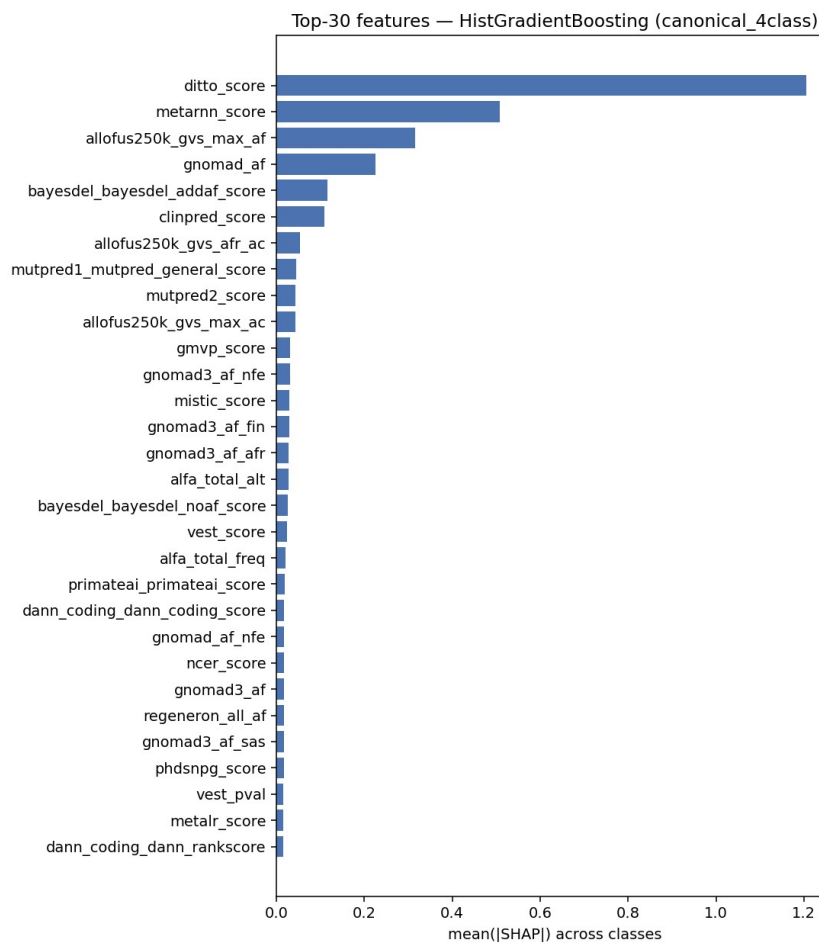
# 5-class confusion matrix — where VUS sits

Held-out, row-normalized. VUS row reaches 0.94 diagonal — cleaner than Pathogenic (0.64) or Likely-pathogenic (0.61). VUS misclassifications hedge into Likely-\*, not into definitive calls.



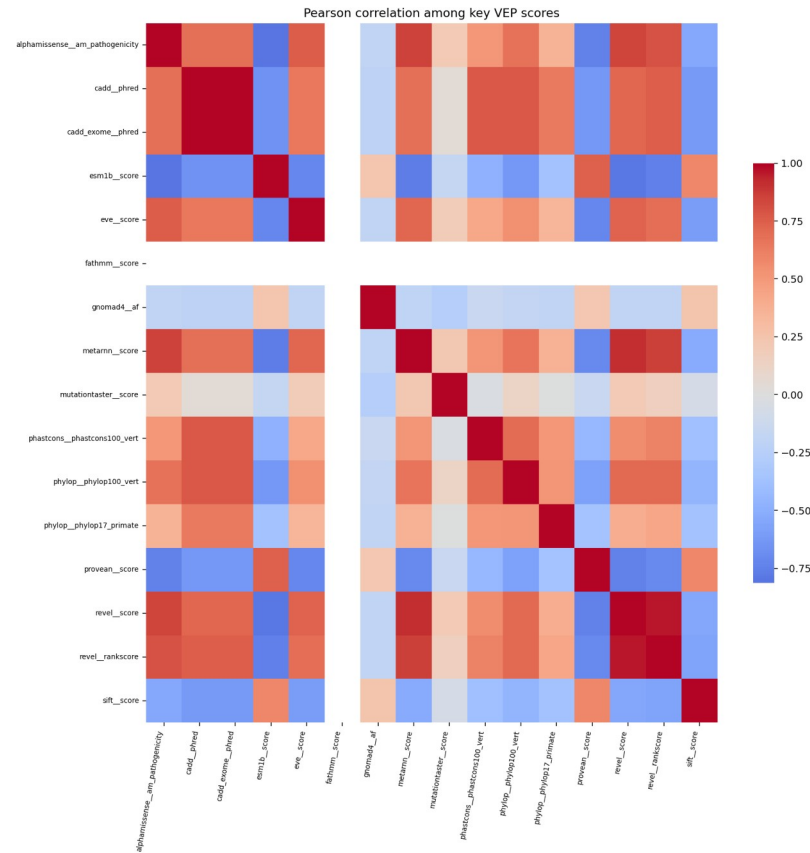
# What drives the booster — SHAP top-30

*Canonical 4-class HistGradientBoosting. DITTO leads, MetaRNN second, then population allele frequencies — the model is using both predictor scores and frequency data.*



# Predictor correlation — the redundancy structure

*Pearson correlation among key VEP scores + gnomAD AF. Predictor scores form a tight cluster; gnomAD AF sits in its own corner — the structural reason predictors substitute for each other, but population AF does not.*





# Useful vs. waste — feature ablation

| Group         | 4-class | 2-class | 5-class | 3-class | n feat |
|---------------|---------|---------|---------|---------|--------|
| population_af | +0.0471 | +0.0025 | +0.0405 | +0.0054 | 47     |
| functional    | -0.0012 | -0.0005 | +0.0266 | +0.0469 | 4      |
| other_vep     | +0.0133 | -0.0002 | +0.0137 | +0.0073 | 56     |
| cadd          | -0.0054 | -0.0005 | -0.0001 | -0.0015 | 4      |
| ditto         | +0.0039 | +0.0046 | +0.0001 | +0.0033 | 1      |
| metarnn       | +0.0014 | +0.0005 | +0.0045 | +0.0005 | 2      |
| bayesdel      | -0.0045 | -0.0002 | -0.0004 | -0.0003 | 4      |
| position      | -0.0044 | +0.0002 | +0.0018 | +0.0029 | 2      |
| revel         | -0.0018 | +0.0005 | -0.0044 | -0.0015 | 2      |
| chasmplus     | +0.0026 | +0.0011 | +0.0028 | -0.0002 | 68     |
| alphamissense | -0.0015 | +0.0005 | -0.0028 | -0.0017 | 1      |
| conservation  | +0.0018 | -0.0007 | -0.0001 | +0.0000 | 17     |

*Greyed rows: HistGB  $|\Delta| < 0.005$  across every task. 8 of 12 groups (97 of 208 features) are waste at the group level.*

# Per-feature permutation — the redundancy story

*Removing a 'waste' group is cheap — but individual features inside those groups are still highly predictive. Group LOO measures replaceability, not informativeness.*

**ditto\_score**

**0.133**

individual macro-F1 weight (4-class)

**metarnn\_score**

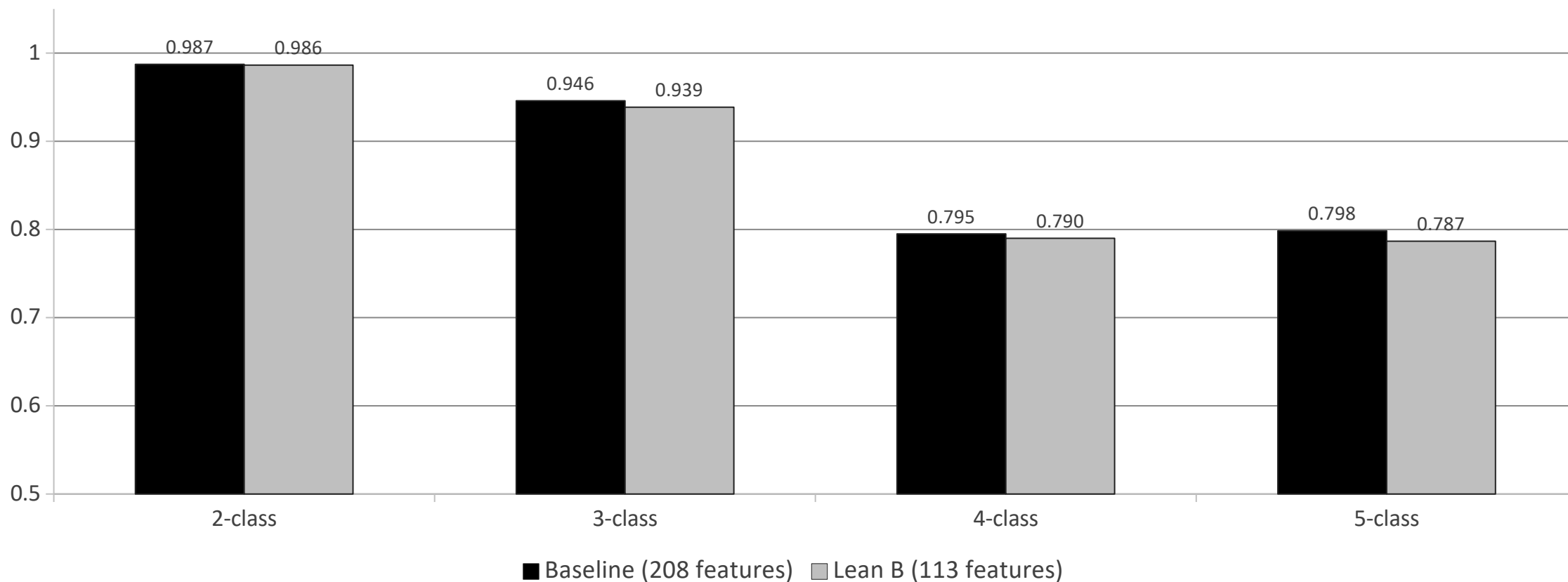
**0.048**

individual macro-F1 weight (4-class)

*Inter-predictor redundancy. Remove DITTO and MetaRNN compensates; remove MetaRNN and DITTO compensates. Only by removing several correlated meta-classifiers together does the redundancy buffer collapse — which is what Lean B exploits.*

# Lean B — 45% smaller, 0.012 cost

*Lean B drops 95 of 208 features, keeping ditto\_score and metarnn\_score (the per-feature winners).*



*At most 0.012 macro-F1 cost on any task. Several models (KNN, NearestCentroid, CosineSimilarity) actually improve on the lean subset.*

# Future directions

- **Cross-database validation**

replicate Lean B numbers on HGMD or a held-out ClinVar snapshot

- **True homology BLAST**

blastp on UniRef50 with the amino-acid substitution applied

- **Probability calibration**

Platt or isotonic regression so predict\_proba is a usable risk score

- **Gene-stratified VUS-as-class**

test whether VUS-detection survives without same-gene leakage

- **Variant-level error analysis**

manual review of the Pathogenic  $\leftrightarrow$  Likely-pathogenic confusion region

# Thank you

*Questions?*

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*[github.com/doruktopcu/MissVarPath-Missense-Detection](https://github.com/doruktopcu/MissVarPath-Missense-Detection)*